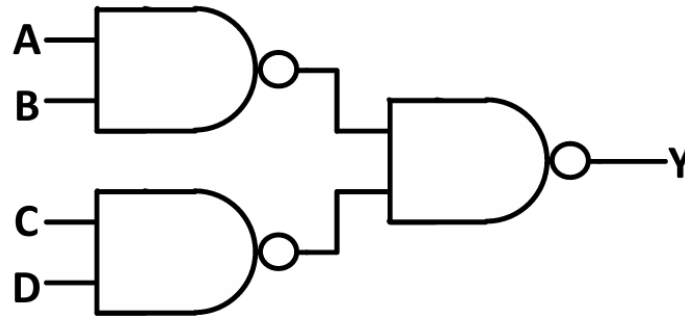


EE 2018 Q-14

Q. In the logic circuit shown in the figure, Y is given by



1. A) $Y = ABCD$
2. B) $Y = (A + B)(C + D)$
3. C) $Y = A + B + C + D$
4. D) $Y = AB + CD$

Boolean Expression Derivation

The given logic circuit consists of three NAND gates.

Step 1: Output of the First Two NAND Gates

The upper NAND gate produces

$$X = (AB)' \quad (1)$$

The lower NAND gate produces

$$W = (CD)' \quad (2)$$

Step 2: Output of the Final NAND Gate

The outputs of the first two NAND gates are connected to a third NAND gate. Therefore,

Truth Table

Table 1 shows the truth table for the Boolean expression $Y = AB + CD$.

Table 1: Truth Table for $Y = AB + CD$

A	B	C	D	AB	CD	Y
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	1	1
0	1	0	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	0	0	0
0	1	1	1	0	1	1
1	0	0	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	0	1	1	0	1	1
1	1	0	0	1	0	1
1	1	0	1	1	0	1
1	1	1	0	1	0	1
1	1	1	1	1	1	1

Minterm Representation

From the truth table, the output is equal to 1 for the following minterms:

$$m(3, 7, 11, 12, 13, 14, 15) \quad (3)$$

Therefore, the canonical Sum-of-Products (SOP) form is

$$Y(A, B, C, D) = \Sigma m(3, 7, 11, 12, 13, 14, 15) \quad (4)$$

Canonical SOP Expression

From the truth table, the output is equal to 1 for the minterms

$$m(3, 7, 11, 12, 13, 14, 15) \quad (5)$$

Therefore,

$$Y(A, B, C, D) = \Sigma m(3, 7, 11, 12, 13, 14, 15) \quad (6)$$

Expanding each minterm,

$$m_3 = A'B'CD \quad m_7 = A'BCD \quad m_{11} = AB'CD \quad m_{12} = ABC'D' \quad m_{13} = ABC'D \quad m_{14} = ABCD' \quad m_{15} = ABCD \quad (7)$$

Hence, the canonical Sum-of-Products (SOP) expression is

$$Y = A'B'CD + A'BCD + AB'CD + ABC'D' + ABC'D + ABCD' + ABCD \quad (8)$$

This expression is obtained directly from the truth table before any Karnaugh Map simplification.

Algebraic Simplification

Starting from the canonical SOP expression,

$$Y = A'B'CD + A'BCD + AB'CD + ABC'D' + ABC'D + ABCD' + ABCD \quad (9)$$

Step 1: Group Terms Having Common Literals

$$Y = (A'B'CD + A'BCD + AB'CD) + (ABC'D' + ABC'D + ABCD' + ABCD) \quad (10)$$

Step 2: Factor (CD) from the First Group

$$Y = CD(A'B' + A'B + AB') + (ABC'D' + ABC'D + ABCD' + ABCD) \quad (11)$$

Step 3: Simplify the First Bracket

Using

$$A'B' + A'B = A'(B' + B) = A' \quad (12)$$

Therefore,

$$A'B' + A'B + AB' = A' + AB' \quad (13)$$

Using

$$X + YZ = (X + Y)(X + Z) \quad (14)$$

$$A' + AB' = (A' + A)(A' + B') \quad (15)$$

$$= 1(A' + B') \quad (16)$$

$$= A' + B' \quad (17)$$

Hence,

$$Y = CD(A' + B') + (ABC'D' + ABC'D + ABCD' + ABCD) \quad (18)$$

Step 4: Factor (AB) from the Remaining Terms

$$Y = CD(A' + B') + AB(C'D' + C'D + CD' + CD) \quad (19)$$

Step 5: Simplify the Second Bracket

$$C'D' + C'D = C'(D' + D) = C' \quad (20)$$

Similarly,

$$CD' + CD = C(D' + D) = C \quad (21)$$

Therefore,

$$C'D' + C'D + CD' + CD = C' + C \quad (22)$$

$$= 1 \quad (23)$$

Hence,

$$Y = CD(A' + B') + AB \quad (24)$$

Step 6: Expand the First Term

$$Y = A'CD + B'CD + AB \quad (25)$$

Step 7: Apply Absorption and Consensus Theorem

Consider

$$Y = AB + A'CD + B'CD \quad (26)$$

Using the Consensus Theorem,

$$XY + X'Z + YZ = XY + X'Z \quad (27)$$

Let

$$X = A, \quad Y = B, \quad Z = CD \quad (28)$$

Then,

$$AB + A'CD + BCD = AB + A'CD \quad (29)$$

Since

$$B'CD + BCD = CD \quad (30)$$

the expression simplifies to

$$AB + CD(A' + B') \quad (31)$$

which reduces to

$$\boxed{Y = AB + CD} \quad (32)$$

Karnaugh Map Simplification

The K-map contains 1's at the minterms

$$m(3, 7, 11, 12, 13, 14, 15) \quad (33)$$

Group 1

Grouping

$$m(12, 13, 14, 15) \quad (34)$$

gives

$$AB \quad (35)$$

Group 2

Grouping

$$m(3, 7, 11, 15) \quad (36)$$

gives

$$CD \quad (37)$$

Combining the groups,

$$\boxed{Y = AB + CD} \quad (38)$$

Hardware Connections

Arduino Input Generation

The Arduino Uno is used to generate the four input variables A , B , C , and D automatically.

Table 2: Arduino Input Connections

Arduino Pin	Logic Variable
D9	A
D10	B
D11	C
D12	D

Arduino to SN74LS47N Connections

The output generated by Arduino is applied to the SN74LS47N BCD-to-Seven-Segment Decoder.

Table 3: Arduino to SN74LS47N Connections

Arduino Pin	SN74LS47N Input
D2	A
D3	B
D4	C
D5	D

SN74LS47N to Seven-Segment Connections

The segment outputs of the SN74LS47N are connected to the corresponding segments of the Common Anode Seven-Segment Display.

Table 4: SN74LS47N to Seven-Segment Connections

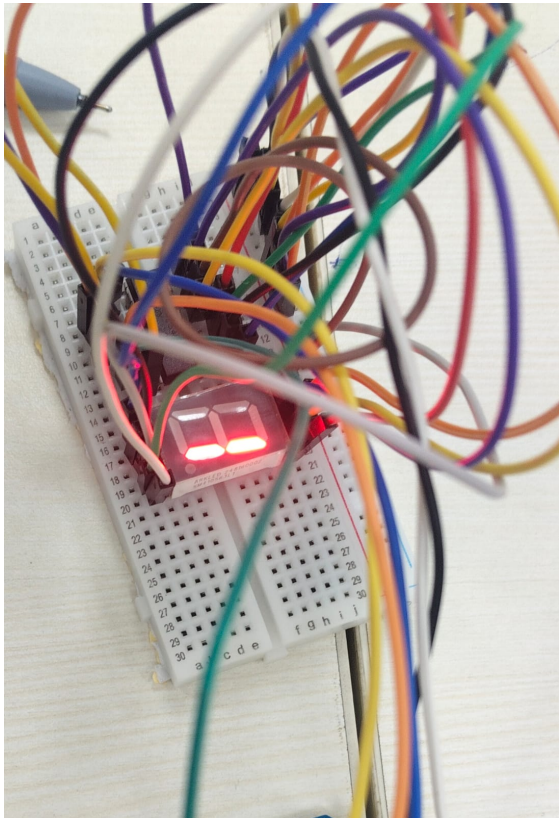
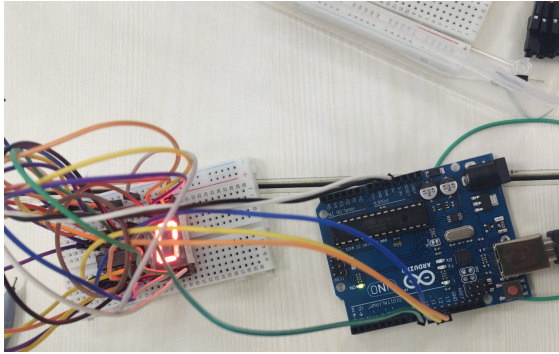
SN74LS47N Output	Seven-Segment Pin
a	Segment a
b	Segment b
c	Segment c
d	Segment d
e	Segment e
f	Segment f
g	Segment g

Note: The common anode terminal of the seven-segment display is connected to $+5V$.

Result

The Boolean expression obtained from the given NAND gate circuit was simplified using the Karnaugh Map technique as

$$\boxed{Y = AB + CD} \quad (39)$$



The simplified expression was successfully implemented using Arduino Uno, SN74LS47N, and a Common Anode Seven-Segment Display. The output values of Y were correctly displayed on the seven-segment display for all sixteen possible input combinations.